1 Model Setup

1.1 Stochastic Primitives

- B_f: persistent fundamental Brownian
- B_s: persistent belief Brownian ($B_f \perp B_s$)
- $\{B_\epsilon^{(j)}\}_{j=1}^J$: idiosyncratic Brownians, one per asset
- $\xi_f \in \mathbb{R}^{n_f}$: auxiliary fundamental shock vector
- $\xi_s \in \mathbb{R}^{n_s}$: auxiliary belief shock vector

1.2 Belief Structure

Scalar belief wedge $s(t)$ acts on B_f via Girsanov

$$d\tilde{B}_f = dB_f - s(t)dt$$

Two specifications for $s(t)$ dynamics

1. DSSW specification
$$ds(t) = -\psi_s\,(s(t) - \bar{s})\,dt + \sigma_s dB_s(t)$$
2. MN specification
$$ds(t) = -\psi_m s(t)dt + \zeta_m dB_f(t)$$

1.3 Production Environemnts

1. Lucas Tree Endowment
$$Y(t) = D(t), \qquad \frac{dD(t)}{D(t)} = \mu_D dt + \sigma_D dB_f(t)$$
2. Brunnermeier-Sannikov AK
$$Y(t) = \bar{A}K(t), \qquad \frac{dK(t)}{K(t)} = (\iota(t) - \delta)dt + \sigma_K dB_f(t)$$
with investment ι chosen by representitive firm FOC (Tobin's q channel). Convex adjustment cost $\frac{\phi_\iota}{2}(\iota - \delta)^2$

For notational convenience, we define the scaling variable

$$X(t) = \begin{cases} D(t) & \text{for endowment} \\ K(t) & \text{for production} \end{cases}$$

with dynamics $\frac{dX(t)}{X(t)} = \mu_X(t)dt + \sigma_X dB_f(t)$. Then,

$$\mu_X(t) = \begin{cases} \mu_D & \text{endowment} \\ (\iota(t) - \delta) & \text{production} \end{cases} \quad \text{and,} \quad \sigma_X = \begin{cases} \sigma_D & \text{endowment} \\ \sigma_K & \text{production} \end{cases}$$

1.4 Demographics

- Continuum of agents (unit mass)
- Type composition: α type $\mathcal{N}$ (noise) and $(1-\alpha)$ type $\mathcal{R}$ (rational)
- Poisson exit rate λ (Blanchard-Yaari Perpetual youth)
- Annuity market: deceased wealth accrues to survivors
- Newborns enter with endowment $W_e(t) = \tau Y(t)/\lambda$.

1.5 Asset Menu

1. Risk-free bond, in zero net supply, paying an instantaneous risk-free rate $r(t)$ determined in equilibrium.
2. Aggregate Market Portfolio. claim on the aggregate payout in positive net supply (1 unit)

$$dR^{(M)}(t) = \frac{dP^{(M)}(t) + d\Pi^{(M)}(t)}{P^{(M)}(t)}$$

$$\text{where } d\Pi^{(M)}(t) = \begin{cases} ((1-\tau)D(t))\,dt & \text{endowment} \\ \left(\left[\bar{A}(1-\tau) - \iota - \frac{\phi_\iota}{2}(\iota-\delta)^2\right]K(t)\right)dt & \text{production} \end{cases}$$

3. J Risky Assets. J sectoral claims with primitive fundamental loadings $\mathbf{b}^{(j)}$ (scalar)
   - Each sector j has supply weight $\chi^{(j)} > 0$ with $\sum_j \chi^{(j)} = 1$. The cross-sectional constraint:

$$\sum_j \chi^{(j)}\mathbf{b}^{(j)} = 1$$

     (sectoral aggregates equals market loading)
   - Primitive payoff for sector j

$$d\Pi^{(j)}(t) = \chi^{(j)}\Pi^{(M)}(t)dt + \chi^{(j)}\mathbf{b}^{(j)}\sigma_X X(t)dB_f(t) + \chi^{(j)}\sigma_\epsilon^{(j)}X(t)dB_\epsilon^{(j)}(t)$$

1.6 Equilibrium Return Process

We let $Z = (s, \omega) \in (\mathbb{R} \times [0,1])$ be the aggregate state vector. Its generalized Itô dynamics are

$$dZ = \mu_Z dt + \Sigma_Z d\mathbf{B}_Z$$

where

$$\begin{cases} d\mathbf{B}_Z = \begin{bmatrix} dB_f \\ dB_s \end{bmatrix}, \quad \Sigma_Z = \begin{bmatrix} 0 & \sigma_s \\ \sigma_f^\omega & \sigma_s^\omega \end{bmatrix} & \text{(for DSSW)} \\ d\mathbf{B}_Z = dB_f, \quad \Sigma_Z = \begin{bmatrix} \zeta_m \\ \sigma_f^\omega \end{bmatrix} & \text{(for MN)} \end{cases}$$

1. The Market Portfolio Return:

$$dR^{(M)}(t) = \mu_R^{(M)}(Z,t)dt + \sigma_f^{(M)}(Z,t)dB_f + \sigma_s^{(M)}(Z,t)dB_s.$$

2. Sector j asset Return:

$$dR^{(j)}(t) = \mu_R^{(j)}(Z,t)dt + \sigma_f^{(j)}(Z,t)dB_f + \sigma_s^{(j)}(Z,t)dB_s + \sigma_\epsilon^{(j)}dB_\epsilon^{(j)}.$$

- For a tractable cross-section, we make the following structural assumptions

$$\sigma_f^{(j)} = \mathbf{b}^{(j)} g_f(Z), \qquad \sigma_s^{(j)} = \mathbf{b}^{(j)} g_s(Z)$$

- Let

$$\Sigma_f = \begin{bmatrix} \sigma_f^{(M)} \\ \sigma_f^{(1)} \\ \sigma_f^{(2)} \\ \vdots \\ \sigma_f^{(J)} \end{bmatrix}, \quad \Sigma_s = \begin{bmatrix} \sigma_s^{(M)} \\ \sigma_s^{(1)} \\ \sigma_s^{(2)} \\ \vdots \\ \sigma_s^{(J)} \end{bmatrix}, \quad \text{and } \Sigma_R = \begin{bmatrix} \Sigma_f & \Sigma_R \end{bmatrix}$$

1.7 Wealth Dynamics

Let C be consumption, and $\Theta = \left(\theta^{(M)}, \theta^{(1)}, \ldots, \theta^{(J)}\right)$ be dollar values invested in the risky assets. The wealth dynamics:

$$\begin{aligned} dW =& \theta^{(M)} dR^{(M)} + \sum_j \theta^{(j)} dR^{(j)} + \left(W - \theta^{(M)} - \sum_j \theta^{(j)}\right) r dt - C dt + \lambda W dt \\ \implies \frac{dW}{W} =& \pi^{(M)} dR^{(M)} + \sum_j \pi^{(j)} dR^{(j)} + \left(1 - \pi^{(M)} - \sum_j \pi^{(j)}\right) r dt - c dt + \lambda dt \end{aligned}$$

where $c \equiv C/W$ (consumption rate) and $\pi = \left(\pi^{(M)}, \pi^{(1)}, \ldots, \pi^{(J)}\right)$ are the portfolio shares.

1.8 Agent HJB

1.8.1 HJB

(In the Baseline) We have representative agents for both type $\mathcal{R}$ and type $\mathcal{N}$. Letting $k \in \{\mathcal{N}, \mathcal{R}\}$

$$(\rho + \lambda) V^k(W, Z) = \max_{c,\theta} \left\{ \frac{C^{1-\gamma}}{1-\gamma} + \mathcal{L}^k V^k \right\}$$

where $\mathcal{L}^k$ is the Itô generater (under $\mathbb{P}$ for $k = \mathcal{R}$ and $\tilde{\mathbb{P}}$ for $k = \mathcal{N}$). We define the objective expected return vector (μ_e) and the expected returns for each type-k agent (μ_e^k):

$$\mu_e = \underbrace{\begin{bmatrix} \mu_R^{(M)} - r \\ \mu_R^{(1)} - r \\ \mu_R^{(2)} - r \\ \vdots \\ \mu_R^{(J)} - r \end{bmatrix}}_{(J+1)\times 1}, \qquad \mu_e^k = \begin{cases} \mu_e & \text{(for } k = \mathcal{R}) \\ \mu_e + s \cdot \Sigma_f & \text{(for } k = \mathcal{N}) \end{cases},$$

Then, the **subjective optimization constraint** (wealth dynamics under type k agent's measure):

$$\frac{dW}{W} = \left[r - c + \lambda + \pi^\top \mu_e^k\right] dt + \pi^\top \Sigma_R d\mathbf{B}_Z^k + \pi^\top \Sigma_\epsilon d\mathbf{B}_\epsilon.$$

where

$$d\mathbf{B}_Z^k = \begin{cases} \begin{bmatrix} dB_f^k \\ dB_s \end{bmatrix} & \text{(for DSSW)} \\ dB_f^k & \text{(for MN)} \end{cases}, \quad d\mathbf{B}_\epsilon = \begin{bmatrix} dB_\epsilon^{(1)} \\ dB_\epsilon^{(2)} \\ \vdots \\ dB_\epsilon^{(J)} \end{bmatrix}_{J\times 1}, \quad \Sigma_\epsilon = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ \sigma_\epsilon^{(1)} & 0 & \cdots & 0 \\ 0 & \sigma_\epsilon^{(2)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_\epsilon^{(J)} \end{bmatrix}_{(J+1)\times J}$$

Finally, letting $\Omega_{(J+1)\times(J+1)} = \Sigma_R \Sigma_R^\top + \Sigma_\epsilon \Sigma_\epsilon^\top$, the HJB becomes:

$$\begin{aligned}(\rho+\lambda)V^k = \max_{c,\pi} \Bigg\{ & \frac{(cW)^{1-\gamma}}{1-\gamma} + V_W^k W(r - c + \lambda + \pi^\top \mu_e^k) \\ & + \frac{1}{2} V_{WW}^k W^2 (\pi^\top \Omega \pi) + \mathcal{L}_Z^k V^k + \mathrm{Cov}^k\left(\frac{dW}{W}, dZ\right) W V_{WZ}^k \Bigg\}.\end{aligned}$$

1.8.2 Reduced HJB

With CRRA preferences with linear wealth dynamics (as in 1.7)

$$V^k(W,Z) = \frac{W^{1-\gamma}}{1-\gamma} v^k(Z)$$

where $v^k(Z)$ is the normalized value function depending only on the aggregate state Z and not individual wealth. Then substituting

$$V_W^k = W^{-\gamma} v^k, \qquad V_{WW}^k = -\gamma W^{-\gamma-1} v^k$$

into the HJB yields

$$\begin{aligned}(c^*)^k &= (v^k)^{-\frac{1}{\gamma}} \\ (\pi^*)^k &= \frac{1}{\gamma}\Omega^{-1}\left(\mu_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_Z v^k}{v^k}\right)\end{aligned}$$

where $\nabla_Z v^k = \begin{bmatrix} \partial_s v^k \\ \partial_\omega v^k \end{bmatrix}$. So the final (simplified) reduced for agent of type-k becomes:

$$0 = \gamma (v^k)^{\frac{\gamma-1}{\gamma}} + v^k\left[(r-\rho) - \gamma(r+\lambda)\right] + \frac{(1-\gamma)v^k}{2\gamma}\left\|\mu_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_Z v^k}{v^k}\right\|_{\Omega^{-1}}^2 + \mathcal{L}_z^k v^k,$$

where $\mathcal{L}_Z^k$ is the Itô generator acting strictly on the state vector, and $\|x\|_{\Omega^{-1}}^2 := x^\top \Omega^{-1} x$ is the Mahalanobis-like norm.

2 Auxiliary Shocks

2.1 Model Setup with Auxiliary Shocks

2.1.1 Auxiliary Shock Definition

We define auxiliary shocks to have vector fast-OU processes. Letting $i \in \{f, s\}$

$$d\xi_i(t) = -\frac{1}{\varepsilon}\xi_i(t)dt + \frac{1}{\sqrt{\varepsilon}}\Sigma_i^{1/2} d\mathbf{B}_i^{\xi}(t)$$

where:

- ε is a small parameter such that $\frac{1}{\varepsilon}$ is the fast decay rate
  - $\varepsilon \to 0$ corresponds to faster reversion
  - *(c.f.)* $1/\sqrt{\varepsilon}$ is the right scaling for Talor expansion to be informative – as $\varepsilon \to 0$, the shocks become *i.i.d.*-like (no temporal correlation) but maintain finite spatial variance.
- $\Sigma_k \in \mathbb{R}^{n_k \times n_k}$ is the covariance structure of the short-lived fundamental (belief) shocks
- $\mathbf{B}_k^{\xi}$ are vector Brownian motions independent of B_f and B_s

2.1.2 Effect on the Production Environment

ξ_f enters as a *small* perturbation to drift.

$$\frac{dX(t)}{X(t)} = \mu_X(t)dt + \sigma_X dB_f(t) + \left(\mathbf{a}_X \cdot \xi_f(t)\right) dt$$

where $\mathbf{a}_X \in \mathbb{R}^{n_f}$ are the loadings of aggregate scaling on auxiliary fundamental shocks

2.1.3 Effect on Equilibrium Return Processes

$$dR^{(j)}(t) = \mu_R^{(j)} dt + \sigma_f^{(j)} dB_f + \sigma_s^{(j)} dB_s + \left(\mathbf{a}_f^{(j)} \cdot \xi_f\right) dt + \left(\mathbf{a}_s^{(j)} \cdot \xi_s\right) dt + \sigma_\epsilon^{(j)} dB_\epsilon^{(j)}$$

where asset j's primitive loading on each auxiliary shock dimension are denoted $\mathbf{a}_f^{(j)} \in \mathbb{R}^{n_f}$ and $\mathbf{a}_s^{(j)} \in \mathbb{R}^{n_s}$. The cross-sectional constraints:

$$\sum_j \chi^{(j)} \mathbf{a}_f^{(j)} = \mathbf{a}_X, \qquad \sum_j \chi^{(j)} \mathbf{a}_s^{(j)} = \mathbf{0}.$$

i.e. Sectoral aggregates of fundamental auxiliary loadings equal aggregate loadings; sectoral aggregates of belief auxiliary loadings sum to zero because the market portfolio doesn't load on belief shocks at the primitive level. We define the auxiliary shock matrices:

$$A_f = \begin{bmatrix} (a_f^{(M)})^\top \\ (a_f^{(1)})^\top \\ \vdots \\ (a_f^{(J)})^\top \end{bmatrix}_{(J+1)\times n_f} \qquad A_s = \begin{bmatrix} \mathbf{0}_{n_s}^\top \\ (a_f^{(1)})^\top \\ \vdots \\ (a_f^{(J)})^\top \end{bmatrix}_{(J+1)\times n_s}$$

where the top row of A_s is 0 because the market portfolio does not load on belief shocks.

2.1.4 Effect on HJB

The (reduced) HJB now includes terms for ξ_f, ξ_s dynamics. Letting $\hat{Z} = (Z, \xi_f, \xi_s)$

$$(\rho + \lambda)v^k(\hat{Z}) = \max_{c,\pi} \left\{ u(c) + \hat{\mathcal{L}}^k v^k \right\}$$

where the expanded generator $\hat{\mathcal{L}}^k$ now includes ξ_f, ξ_s-derivatives and cross-derivatives between ξ_f, ξ_s and other states:

$$\hat{\mathcal{L}}^k v^k = \mathcal{L}_Z^k v^k + \sum_{i \in \{f,s\}} \left[-\frac{1}{\varepsilon} \left(\xi_i \cdot \nabla_{\xi_i} v^k \right) + \frac{1}{2\varepsilon} \mathrm{tr} \left(\Sigma_i \nabla^2_{\xi_i} v^k \right) \right].$$

(Note. because the Brownians are independent, there are no cross-partial derivative terms between Z and ξ, nor between ξ_f and ξ_s). We define the *expanded* excess return vector for type-k agents that include the linear perturbations

$$\hat{\mu}_e^k = \mu_e^k + A_f \xi_f + A_s \xi_s.$$

Then, the ***expanded* subjective optimization constraint** (*expanded* wealth dynamics under type k agent's measure):

$$\frac{dW}{W} = \left[r - c + \lambda + \pi^\top \hat{\mu}_e^k \right] dt + \pi^\top \Sigma_R dB_Z^k + \pi^\top \Sigma_\epsilon dB_\epsilon.$$

Due to our assumptions of $d\mathbf{B}_i^\xi \perp d\mathbf{B}_Z^k$ and $d\mathbf{B}_i^\xi \perp d\mathbf{B}_\epsilon$ (for $i \in \{f, s\}$),

$$\mathrm{Cov}^k \left(\frac{dW}{W}, d\xi_f \right) = 0, \qquad \mathrm{Cov}^k \left(\frac{dW}{W}, d\xi_s \right) = \mathbf{0}$$

$$\Longrightarrow \begin{cases} (c^*)^k = & (v^k)^{-\frac{1}{\gamma}} \\ (\pi^*)^k = & \frac{1}{\gamma} \Omega^{-1} \left(\hat{\mu}_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_Z v^k}{v^k} \right) \end{cases}$$

where the gradient terms $\nabla_{\xi_f} v^k$ and $\nabla_{\xi_s} v^k$ are missing from $(\pi^*)^k$ because the local risk cannot be traded. The HJB thus becomes

$$\begin{aligned} 0 =& \gamma (v^k)^{\frac{\gamma-1}{\gamma}} + v^k \left[(r - \rho) - \gamma (r + \lambda) \right] + \frac{(1-\gamma) v^k}{2\gamma} \left\| \hat{\mu}_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_Z v^k}{v^k} \right\|^2_{\Omega^{-1}} + \mathcal{L}_z^k v^k \\ &+ \sum_{i=\{f,s\}} \left[-\frac{1}{\varepsilon} \left(\xi_i \cdot \nabla_{\xi_i} v^k \right) + \frac{1}{2\varepsilon} \mathrm{tr} \left(\Sigma_i \nabla^2_{\xi_i} v^k \right) \right] \end{aligned}$$

2.2 Taylor Expansion

With the expanded state space $\hat{Z}$, the full state vector has $2 + n_f + n_s$ dimensions. We want to expand the value function as

$$v^k(\hat{Z}) = v_0^k(Z) + \varepsilon v_1^k(\hat{Z})$$

where the first term $v_0^k(Z)$ captures the persistent, macroeconomic risk and the second term $\varepsilon v_1^k(\hat{Z})$ isolates the transient, unhedgeable background risk. Because the auxiliary ξ shocks mean-revert so violently as $\varepsilon \to 0$, the expansion proves that agents do not dynamically hedge against them; they only react to them *myopically* as temporary shifts in the expected return drift. Mathematically we want to show that we only need to solve a 2D nonlinear PDE for $v_0^k(Z)$, while $\varepsilon v_1^k(\hat{Z})$ can be solved using tractable linear algebra.

2.2.1 Taylor Expansion of HJB

We plug the conjecture $v^k = v_0^k + \varepsilon v_1^k + O(\varepsilon^2)$ into the fully expanded HJB block by block:

$$0 = \underbrace{\gamma (v^k)^{\frac{\gamma-1}{\gamma}}}_{\text{Block 1}} + \underbrace{v^k \left[(r-\rho) - \gamma(r+\lambda)\right]}_{\text{Block 2}} + \underbrace{\frac{(1-\gamma)v^k}{2\gamma} \left\| \hat{\mu}_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_Z v^k}{v^k} \right\|_{\Omega^{-1}}^2}_{\text{Block 3}} + \underbrace{\mathcal{L}_z^k v^k}_{\text{Block 4}}$$
$$+ \underbrace{\sum_{i=\{f,s\}} \left[-\frac{1}{\varepsilon} \left(\xi_i \cdot \nabla_{\xi_i} v^k \right) + \frac{1}{2\varepsilon} \mathrm{tr} \left(\Sigma_i \nabla_{\xi_i}^2 v^k \right) \right]}_{\text{Block 5}}$$

- **[Block 1] The Utility Term.** Using standard Taylor approximation $(x + \Delta x)^n \approx x^n + nx^{n-1}\Delta x$:

$$\gamma (v_0^k + \varepsilon v_1^k + O(\varepsilon^2))^{\frac{\gamma-1}{\gamma}} = \boxed{\gamma (v_0)^{\frac{\gamma-1}{\gamma}} + \varepsilon(\gamma-1)(v_0^k)^{-\frac{1}{\gamma}} v_1^k + O(\varepsilon^2)}$$

- **[Block 2] The Linear Drift.** Define $\bar{\rho} \equiv [(r-\rho) - \gamma(r+\lambda)]$ as the effective discount rate for notational convenience

$$\bar{\rho}(v_0^k + \varepsilon v_1^k + O(\varepsilon^2)) = \boxed{\bar{\rho} v_0^k + \varepsilon \bar{\rho} v_1^k + O(\varepsilon^2)}$$

- **[Block 3] The Mahalanobis Norm Term.** This term is highly non-linear, so we use the Gâteaux Approach. Temporarily define this block as a single, non-linear operator $\mathcal{F}$:

$$\mathcal{F}(v, \xi) \equiv \frac{(1-\gamma)v}{2\gamma} \left\| \hat{\mu}_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_Z v}{v} \right\|_{\Omega^{-1}}^2$$

  and a scalar function $\mathcal{G}$ that tracks the output of $\mathcal{F}$ as we vary ε:

$$\mathcal{G}(\varepsilon) = \mathcal{F}(v_0^k + \varepsilon v_1^k, \xi).$$

  Then because $\mathcal{F}$ is smooth and differentiable, we can write the standard Maclaurin series

$$\mathcal{G}(\varepsilon) = \mathcal{G}(0) + \varepsilon \mathcal{G}'(0) + O(\varepsilon^2)$$
$$\text{where} \begin{cases} \mathcal{G}(0) & = \mathcal{F}(v_0^k, \xi) \\ \mathcal{G}'(0) & = \frac{d}{d\varepsilon} \mathcal{F}(v_0^k + \varepsilon v_1^k, \xi)\big|_{\varepsilon=0} \equiv \mathcal{M}_1 \end{cases}$$

  So writing it out in full [Block 3] becomes:

$$\frac{(1-\gamma)v^k}{2\gamma} \left\| \hat{\mu}_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_Z v^k}{v^k} \right\|_{\Omega^{-1}}^2 = \boxed{\frac{(1-\gamma)v_0^k}{2\gamma} \left\| \hat{\mu}_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_Z v_0^k}{v_0^k} \right\|_{\Omega^{-1}}^2 + \varepsilon \mathcal{M}_1 + O(\varepsilon^2)}$$

  where $\mathcal{M}_1 = \frac{d}{d\varepsilon} \mathcal{F}(v_0^k + \varepsilon v_1^k, \xi)\big|_{\varepsilon=0}$

- **[Block 4] The Slow Generator $\mathcal{L}_Z^k$.**

$$\mathcal{L}_Z^k \left(v_0^k + \varepsilon v_1^k + O(\varepsilon^2) \right) = \boxed{\mathcal{L}_Z^k v_0^k + \varepsilon \mathcal{L}_Z^k v_1^k + O(\varepsilon^2)}$$

- **[Block 5] The Fast Generator** $\mathcal{L}_\xi^k \equiv \sum_{i=\{f,s\}} \left[-\left(\xi_i \cdot \nabla_{\xi_i} v^k\right) + \frac{1}{2}\mathbf{tr}\left(\Sigma_i \nabla^2_{\xi_i} v^k\right)\right]$**.** [Block 5] then becomes:

$$\frac{1}{\varepsilon}\mathcal{L}_\xi^k\left(v_0^k + \varepsilon v_1^k + O(\varepsilon^2)\right) = \boxed{\frac{1}{\varepsilon}\mathcal{L}_\xi^k v_0^k + \mathcal{L}_\xi^k v_1^k + O(\varepsilon)}$$

where the resdiual term becomes $O(\varepsilon)$ because $\frac{1}{\varepsilon}$ is multiplied to $O(\varepsilon^2)$.

Combining the 5 blocks and rearranging, the Taylor expanded HJB:

$$0 = \underbrace{\left[\frac{1}{\varepsilon}\mathcal{L}_\xi^k v_0^k\right]}_{O(\varepsilon^{-1})\text{ terms}} + \underbrace{\left[\gamma(v_0^k)^{\frac{\gamma-1}{\gamma}} + v_0^k\bar{\rho} + \mathcal{L}_Z^k v_0^k + \frac{(1-\gamma)v_0^k}{2\gamma}\left\|\hat{\mu}_e^k + \Sigma_R\Sigma_Z^\top\frac{\nabla_z v_0^k}{v_0^k}\right\|^2_{\Omega^{-1}} + \mathcal{L}_\xi^k v_1^k\right]}_{O(1)\text{ terms}}$$
$$+ \underbrace{\varepsilon\left[(\gamma-1)(v_0^k)^{-\frac{1}{\gamma}}v_1^k + \bar{\rho}v_1^k + \mathcal{L}_Z^k v_1^k + \mathcal{M}_1 + O(1)\right]}_{O(\varepsilon)\text{ terms}} + O(\varepsilon^2)$$

2.2.2 Executing The Limits

1. The higher-order residual terms grouped in $O(\varepsilon)$ and $O(\varepsilon^2)$ vanish as $\varepsilon \to 0$.
2. $O(\varepsilon^{-1})$ **(The Singularity)**
   (a) Multiply the entire Taylor-expanded HJB equation by ε and let $\varepsilon \to 0$. The equation becomes

   $$0 = \mathcal{L}_\xi^k v_0^k,$$

   i.e. for any fixed Z, v_0^k must reside in the null space of the fast generator (it is harmonic with respect to the fast shocks).

   (b) Note the two constraints on $\mathcal{L}_\xi^k$ and v_0^k:
      i. $\mathcal{L}_\xi^k$ is the infinitesimal generator of a strictly ergodic, multidimensional Ornstein-Uhlenbeck process.
      ii. The solution for v_0^k must satisfy the standard polynomial growth bounds of CRRA utility.

   $\Longrightarrow$ Due to the Liouville property (ergodic process only permits non-constant harmonic functions if they grow unboundedly), constraints *i.* and *ii.* guarantee the solution is a constant with respect to ξ. Thus:

   $$\nabla_{\xi_f} v_0^k = \mathbf{0}, \qquad \nabla_{\xi_s} v_0^k = \mathbf{0}.$$

   (c) Due to (a) and (b), the term $\mathcal{L}_\xi^k v_0^k$ is identically zero. Substituting this identity back into the original HJB equation deletes the ε^{-1} singularity.

3. **The $\mathcal{O}(1)$ Terms.** (due to *1.* and *2.*) Isolated as $\varepsilon \to 0$:

$$
\begin{aligned}
0 =& \gamma (v_0^k)^{\frac{\gamma-1}{\gamma}} + v_0^k \bar{\rho} + \mathcal{L}_Z^k v_0^k + \frac{(1-\gamma) v_0^k}{2\gamma} \left\| \hat{\mu}_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_z v_0^k}{v_0^k} \right\|_{\Omega^{-1}}^2 + \mathcal{L}_\xi^k v_1^k \\
=& \underbrace{\gamma \, (v_0^k)^{\frac{\gamma-1}{\gamma}} + v_0^k \bar{\rho} + \mathcal{L}_Z^k v_0^k + \frac{(1-\gamma) v_0^k}{2\gamma} \left\| \mu_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_z v_0^k}{v_0^k} \right\|_{\Omega^{-1}}^2}_{\text{HJB without auxiliary shocks: } \mathcal{H}_b(v_0^k)} \\
&+ \underbrace{\frac{(1-\gamma) v_0^k}{\gamma} \left(\mu_e^k + \Sigma_R \Sigma_Z^\top \frac{\nabla_z v_0^k}{v_0^k} \right)^\top \Omega^{-1} (A_f \xi_f + A_s \xi_s)}_{\text{New Forcing Terms (Linear)}} \\
&+ \underbrace{\frac{(1-\gamma) v_0^k}{2\gamma} \left\| A_f \xi_f + A_s \xi_s \right\|_{\Omega^{-1}}^2}_{\text{New Forcing Terms (Quadratic)}} + \underbrace{\mathcal{L}_\xi^k v_1^k}_{\text{Shock Absorber}}
\end{aligned}
$$

For the next section we introduce new notation:

$$
\begin{aligned}
0 &= \mathcal{H}_b(v_0^k) + F_l^k(Z, \xi) + F_q(Z, \xi) + \mathcal{L}_\xi^k v_1^k \\
\text{and } m_e^k(Z) &:= \mu_e^k(Z) + \Sigma_R \Sigma_Z^\top \frac{\nabla z v_0^k(Z)}{v_0^k(Z)}
\end{aligned}
$$

so that

$$
\begin{cases}
F_l^k(Z, \xi) := & \frac{(1-\gamma) v_0^k}{\gamma} (m^k)^\top \Omega^{-1} (A_f \xi_f + A_s \xi_s) \\
F_q(Z, \xi) := & \frac{(1-\gamma) v_0^k}{2\gamma} \left\| A_f \xi_f + A_s \xi_s \right\|_{\Omega^{-1}}^2
\end{cases}.
$$

We make two observations:

- F_l^k, F_q^k are polynomials in (ξ_f, ξ_s) of exactly degree 1 and 2 respectively, with Z dependent coefficients
- The fast generator $\mathcal{L}_\xi^k$ preserves polynomial degree in ξ.

$$
\deg_\xi(\mathcal{L}_\xi^k v_1^k) = \deg_\xi(v_1^k)
$$

Based on the two observations, for the residual equation

$$
F_l^k + F_q + \mathcal{L}_\xi^k v_1^k = -\mathcal{H}_b(v_0^k)
$$

to balance at every polynomial degree in ξ, the function v_1 must be a polynomial of exactly 2 degrees in (ξ_f, ξ_s).

2.2.3 Polynomial Coefficient Matching

We posit the multiplicative form

$$
v_1^k(\hat{Z}) = v_0^k(Z) \cdot \mathcal{P}^k(\hat{Z})
$$

where

$$\mathcal{P}^k(\hat{Z}) = \frac{1}{2}\xi_f^\top H_{ff}^k(Z)\xi_f + \xi_f^\top H_{fs}(Z)\xi_s + \frac{1}{2}\xi_s^\top H_{ss}^k(Z)\xi_s + h_f^k(Z)^\top \xi_f + h_s^k(Z)^\top \xi_s$$

$$\text{with} \begin{cases} H_{ff}^k(Z) \in \mathbb{R}^{(n_f \times n_f)} & \text{(symmetric)} \\ H_{ss}^k(Z) \in \mathbb{R}^{(n_s \times n_s)} & \text{(symmetric)} \\ H_{fs}^k(Z) \in \mathbb{R}^{(n_f \times n_s)} & \\ h_f^k(Z) \in \mathbb{R}^{n_f}, & h_s^k(Z) \in \mathbb{R}^{n_s} \end{cases}$$

Remark. Two remarks on the structure of the conjecture

1. **Multiplicative Scaling.** Since v_0^k depends only on Z,

$$\mathcal{L}_\xi^k(v_1^k) = \mathcal{L}_\xi^k(v_0^k \mathcal{P}^k) = v_0^k \mathcal{L}_\xi^k \mathcal{P}^k$$

2. **Absence of a constant-in-ξ term.** A degree-2 conjecture would in principle include an additive piece $h_0^k(Z)$. We omit this by setting $h_0^k \equiv 0$. This is because an additional $h_0^k(Z)$ would satisfy $\mathcal{L}_\xi^k h_0^k \equiv 0$ trivially and would remain undetermined at this order; it would only be pinned down at $O(\varepsilon)$ by a Fredholm solvability condition.

Applying $\mathcal{L}_\xi^k$ on the conjectured polynomial $\mathcal{P}^k$:

$$\mathcal{L}_\xi^k \mathcal{P}^k = \underbrace{-\left[\xi_f^\top H_{ff}^k \xi_f + 2\xi_f^\top H_{fs}^k \xi_s + \xi_s^\top H_{ss}^k \xi_s\right]}_{\text{degree 2 in } \xi} \underbrace{-\left[(h_f^k)^\top \xi_f + (h_s^k)^\top \xi_s\right]}_{\text{degree 1 in } \xi} + \underbrace{\left[\frac{1}{2}\mathrm{tr}\left(\Sigma_f H_{ff}^k\right) + \frac{1}{2}\mathrm{tr}\left(\Sigma_s H_{ss}^k\right)\right]}_{\text{degree 0 in } \xi}$$

Meanwhile, using $\mathcal{L}_\xi^k \mathcal{P}^k = -\frac{\mathcal{H}_b(v_0^k)}{v_0^k} - \frac{F_l^k}{v_0^k} - \frac{F_q}{v_0^k}$,

$$\begin{aligned} \mathcal{L}_\xi^k \mathcal{P}^k = & \underbrace{-\frac{(1-\gamma)}{2\gamma}\left[\xi_f^\top A_f^\top \Omega^{-1} A_f \xi_f + 2\xi_f^\top A_f^\top \Omega^{-1} A_s \xi_s + \xi_s^\top A_s^\top \Omega^{-1} A_s \xi_s\right]}_{-\frac{F_q}{v_0^k}\text{: degree 2 in } \xi} \\ & \underbrace{-\frac{(1-\gamma)}{\gamma}\left[\left(A_f^\top \Omega^{-1} m_e^k\right)^\top \xi_f + \left(A_s^\top \Omega^{-1} m_e^k\right)^\top \xi_s\right]}_{-\frac{F_l^k}{v_0^k}\text{: degree 1 in } \xi} \\ & \underbrace{-\frac{\mathcal{H}_b(v_0^k)}{v_0^k}}_{\text{degree 0 in } \xi} \end{aligned}$$

Equating the two expressions, letting $i \in \{f, s\}$

$$\begin{aligned} h_i^k(Z) &= \frac{1-\gamma}{\gamma} A_i^\top \Omega^{-1} m_e^k(Z) \\ H_{ii}^k(Z) &= \frac{1-\gamma}{2\gamma} A_i^\top \Omega^{-1} A_i^\top \qquad H_{fs}^k = \frac{1-\gamma}{2\gamma} A_f^\top \Omega^{-1} A_s \end{aligned}$$

Defining

$$\mathbf{H}^k(Z) := \begin{bmatrix} H^k_{ff} & H^k_{fs} \\ (H^k_{fs})^\top & H^k_{ss} \end{bmatrix}_{(n_f+n_s)\times(n_f} \quad \text{, and} \quad \mathbf{A} := \begin{bmatrix} A_f & A_s \end{bmatrix}_{(J+1)\times(n_f+n_s)}$$

$$\Longrightarrow \boxed{\mathbf{H}^k = \frac{1-\gamma}{2\gamma}\mathbf{A}^\top\Omega^{-1}\mathbf{A}}.$$

Matching the degree-0 terms:

$$\mathcal{H}_b(v_0^k) = \frac{(1-\gamma)v_0^k}{4\gamma}\mathrm{tr}\left(\Omega^{-1}\mathbf{A}\Xi\mathbf{A}^\top\right), \quad \text{where } \Xi := \begin{bmatrix} \Sigma_f & 0 \\ 0 & \Sigma_s \end{bmatrix}.$$

2.3 Diagonalization